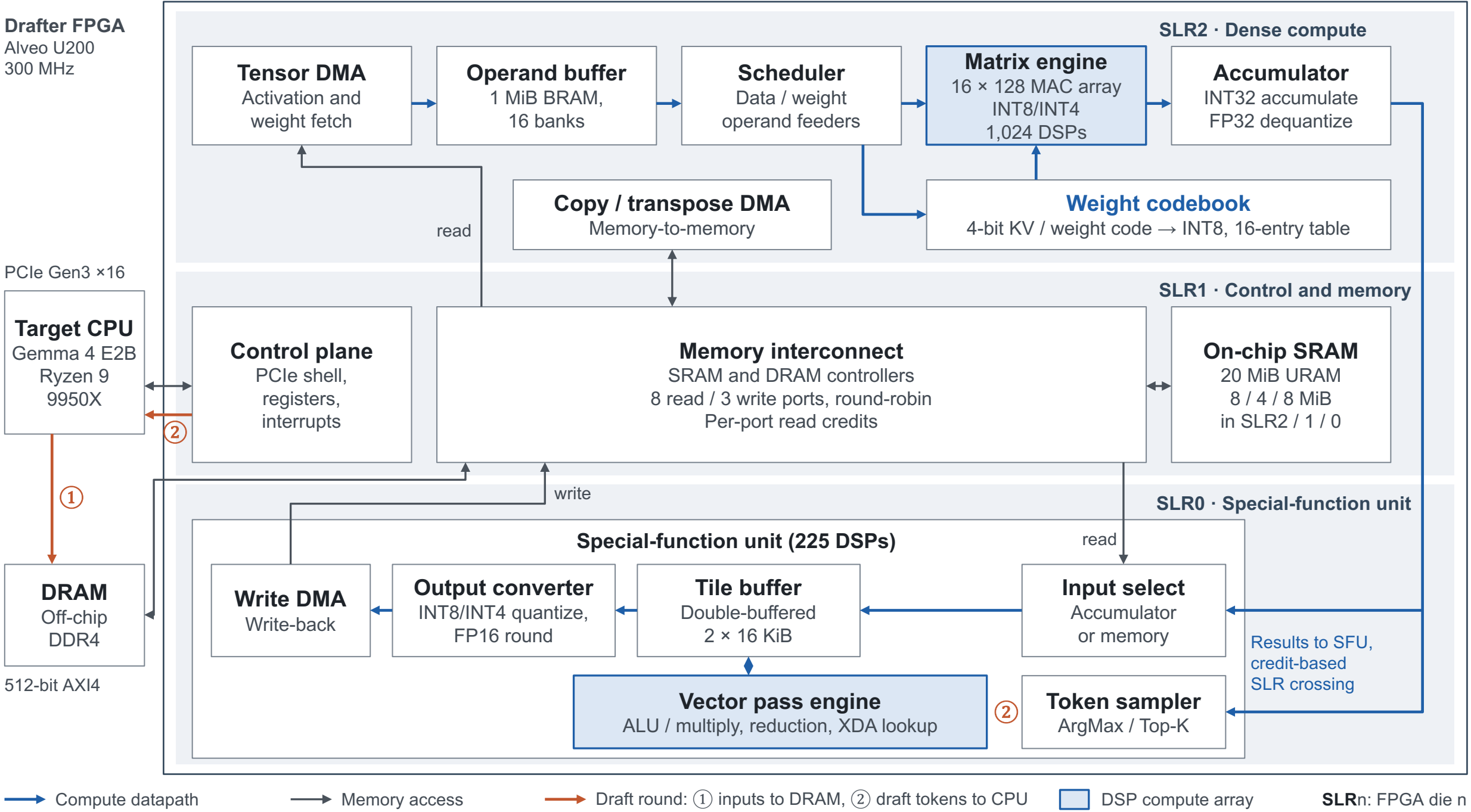
Drafter FPGA
Alveo U200
300 MHz
SLR2 · Dense compute
Tensor DMA
Activation and weight fetch
Operand buffer
1 MiB BRAM, 16 banks
Scheduler
Data / weight operand feeders
Matrix engine
16 × 128 MAC array
INT8/INT4
1,024 DSPs
Accumulator
INT32 accumulate
FP32 dequantize
Copy / transpose DMA
Memory-to-memory
read
Weight codebook
4-bit KV / weight code → INT8, 16-entry table
PCIe Gen3 ×16
SLR1 · Control and memory
Target CPU
Gemma 4 E2B
Ryzen 9
9950X
Control plane
PCIe shell, registers, interrupts
Memory interconnect
SRAM and DRAM controllers
8 read / 3 write ports, round-robin
Per-port read credits
On-chip SRAM
20 MiB URAM
8 / 4 / 8 MiB
in SLR2 / 1 / 0
write
SLR0 · Special-function unit
Special-function unit (225 DSPs)
read
DRAM
Off-chip DDR4
Write DMA
Write-back
Output converter
INT8/INT4 quantize, FP16 round
Tile buffer
Double-buffered
2 × 16 KiB
Input select
Accumulator or memory
Results to SFU, credit-based SLR crossing
Vector pass engine
ALU / multiply, reduction, XDA lookup
Token sampler
ArgMax / Top-K
512-bit AXI4
Compute datapath
Memory access
Draft round: ① inputs to DRAM, ② draft tokens to CPU
DSP compute array
SLRn: FPGA die n